

Exercise 3 from Advanced Econometrics Exam (2014)

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Using data on 133 countries for the period 1970-2009 an economist created unbalanced panel data set. The economist estimated an model to explain differences in life expectancy at birth among countries. The dependent variable was life expectancy at birth in years. The covariates were:

1. functional **literacy** as a percent of population that can read
2. GDP per capita in US dollars
3. time trend to catch the effect of progress in medicine
4. *post_comm* – a dummy variable that takes values 1 for countries that used to be or still is a communist country part. With this variable the economist wanted to verify the hypothesis that in communist countries had relatively good medical care.

The model was estimated using OLS method, fixed effects and random effects estimator. The results are presented below. Use 5% significance level. All the answers should be thoroughly justified.

- a) Explain why robust estimator was used with OLS method.
- b) Compare the results from OLS and fixed effects model only. Which one is more effective?
- c) Using the results choose the most appropriate estimator between POLS, FE, and RE. Explain in what is the null hypothesis we test?
- d) Why is there no estimate for *post_comm* in fixed effects model?
- e) Interpret all R^2 statistics for fixed effects model.
- f) Should we omit individual effects in the fixed effects estimator?
- g) Why did we received in RE and FE models two estimates for random factors (σ_u , σ_e)?

Linear regression

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
life							
lit		.3069304	.0160516	19.12	0.000	.2751788	.338682
GDP_pc		.0002563	.0000399	6.43	0.000	.0001775	.0003352
post_comm		-1.302394	.8825781	-1.48	0.142	-3.048221	.4434327
year		-.0098472	.0260935	-0.38	0.706	-.0614626	.0417683
_cons		60.10586	51.21889	1.17	0.243	-41.21018	161.4219

Random-effects GLS regression

life	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
lit	.1054896	.007226	14.60	0.000	.0913269	.1196522
GDP_pc	.0000376	7.16e-06	5.24	0.000	.0000235	.0000516
post_comm	1.562903	1.079774	1.45	0.148	-.5534152	3.679221
year	.1900848	.0068268	27.84	0.000	.1767046	.2034649
_cons	-321.9032	13.15153	-24.48	0.000	-347.6797	-296.1266
sigma_u	4.4966449					
sigma_e	2.5231168					
rho	.76054561	(fraction of variance due to u_i)				

Breusch and Pagan Lagrangian multiplier test for random effects

chi2(1) = 43672.49

Prob > chi2 = 0.0000

Fixed-effects (within) regression

R-sq: within = 0.5793

between = 0.4524

overall = 0.2970

life	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
lit	.0468977	.0078084	6.01	0.000	.0315894	.062206
GDP_pc	-2.12e-07	7.28e-06	-0.03	0.977	-.0000145	.0000141
post_comm	(omitted)					
year	.2384537	.0071899	33.17	0.000	.2243581	.2525493
_cons	-413.1269	13.79678	-29.94	0.000	-440.1753	-386.0785
sigma_u	8.8759641					
sigma_e	2.5231168					
rho	.92523542	(fraction of variance due to u_i)				

F test that all u_i=0: F(132, 4533) = 127.85 Prob > F = 0.0000

Hausman test

chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B) = 301.07

Prob>chi2 = 0.0000